

Biomaterials

生物材料

南方科技大学材料科学与工程系2024年春季课程

Degradation of Materials in the Biological Environment

A: Introduction

B: Chemical and biochemical degradation of polymers

C: Degradative effects of the biological environment on metals and ceramics

Introduction

The biological environment is surprisingly harsh and can lead to rapid or gradual breakdown of many materials.

Superficially, one might think that the neutral pH, low salt content, and modest temperature of the body would constitute a mild environment.

However, many special mechanisms are brought to bear on implants to break them down. These are mechanisms that have evolved over millennia specifically to rid the living organism of invading foreign substances and they now attack contemporary biomaterials.

Introduction

To understand the biological degradation of implant materials, synergistic pathways should be considered.

For example;

- (1) The **cracks** associated with stress crazing open up fresh surface area to reaction.
- (2) **Swelling and water uptake** can similarly increase the number of sites for reaction.
- (3) **Degradation products** alter the local pH, stimulating further reaction.
- (4) **Hydrolysis of polymers** can generate more hydrophilic species, leading to polymer swelling and entry of degrading species into the bulk of the polymer.
- (5) **Cracks might** also serve as sites for the initiator **calcification**.

Chemical and Biochemical Degradation of Polymers

Biodegradation is the chemical breakdown of materials by the action of living organisms which leads to changes in physical properties.

2-1 Polymer Degradation Processes

No polymer is totally impervious to the chemical processes and mechanical action of the body.

Generally, polymeric biomaterials degrade because

- (1) body constituents attack the biomaterials directly or
- (2) through other device components or
- (3) the intervention of external factors.

TABLE 2 Mechanisms Leading to Degradation of Polymer Properties^a

Physical	Chemical
Sorption	Thermolysis
Swelling	Radical scission
Softening	Depolymerization
Dissolution	<u>Oxidation</u>
Mineralization	Chemical
Extraction	Thermooxidative
Crystallization	Solvolysis
Decrystallization	<u>Hydrolysis</u>
Stress cracking	Alcoholysis
Fatigue fracture	Aminolysis, etc.
Impact fracture	Photolysis
	Visible
	Ultraviolet
	Radiolysis
	Gamma rays
	X-rays
	Electron beam
	Fracture-induced radical reactions

^aSome degradation processes may involve combinations of two or more individual mechanisms.

Hydrolytic Biodegradation

(1) Structure of Hydrolyzable Polymers

Hydrolysis is the scission of susceptible molecular functional groups by reaction with water. It may be catalyzed by acids, bases, salts, or enzymes.

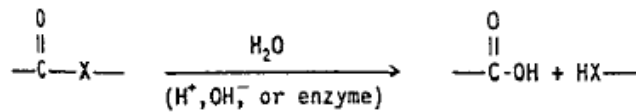
In a commonly used category of hydrolysable polymeric biomaterials, functional groups consist of carbonyls bonded to heterochain elements (O, N, S).

Examples include esters, amides, and carbonates (Fig. 1).

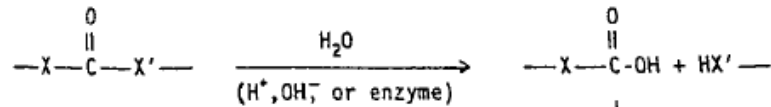
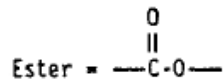
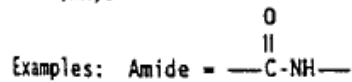
Other polymers containing groups such as ether, acetal, nitrile, phosphonate, or active methylenes, hydrolyze under certain conditions (Fig. 1).

Groups that are normally very stable to hydrolysis are indicated in Fig. 2.

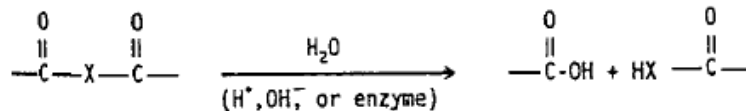
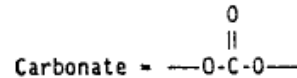
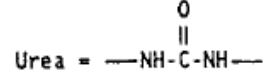
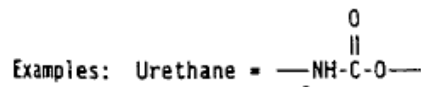
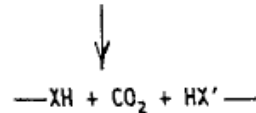
Structure of Hydrolyzable Polymers



X = O, NH, S



X = O, NH, S
X' = O, NH, S



X = O, NH, S

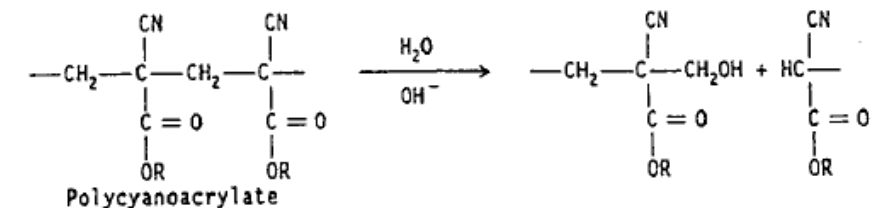
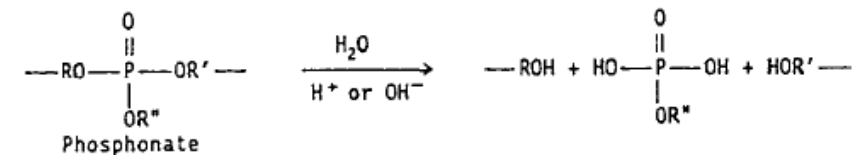
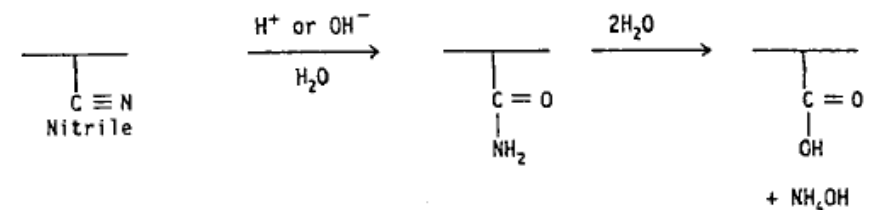
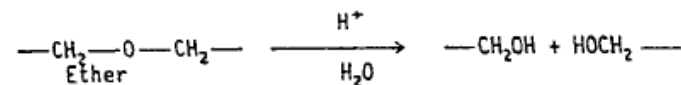
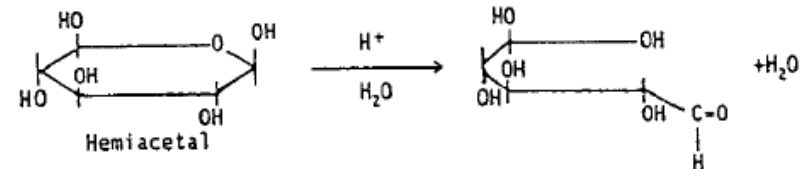
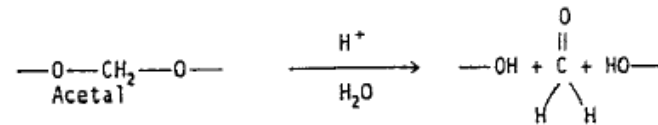
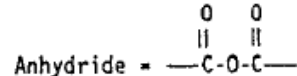
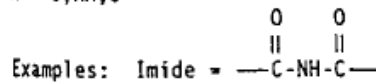
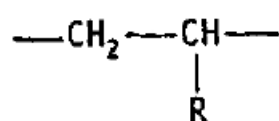


FIG. 1. Hydrolyzable groups in polymer biomaterials.

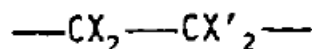
Groups Highly Stable to Hydrolysis



Hydrocarbon

R=H, Alkyl, Aryl

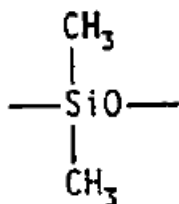
Examples: Polyethylene
Polypropylene
Polystyrene



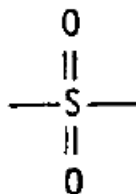
Halocarbon

X=F, Cl, H
X'=F, Cl

Examples: Polytetrafluoroethylene
Polychlorotrifluoroethylene
Polyvinylidene chloride



Dimethylsiloxane



Sulfone

FIG. 2. Groups highly stable to hydrolysis.

Hydrolysis

The rate of hydrolysis tends to increase with a high proportion of hydrolyzable groups in the main or side chain, other polar groups which enhance **hydrophilicity, low crystallinity, low or negligible cross-link density, a high ratio of exposed surface area to volume and, very likely, mechanical stress**. Porous hydrolyzable structures undergo especially rapid property loss because of their large surface area.

Factors that tend to suppress hydrolysis kinetics include hydrophobic moieties (e.g., hydrocarbon or fluorocarbon), cross-linking, high crystallinity due to chain order, thermal annealing or orientation, low stress, and compact shape. While the molecular weight of linear polymers may not have a great effect on degradation rate, physical property losses may be retarded for a given number of chain cleavage events with relatively high-molecular-weight polymers.

Host-induced Hydrolytic Processes

The body is normally a highly controlled reaction medium.

→Through **homeostasis**, the normal environment of most implants is maintained at isothermal (37°C), neutral (pH 7.4), aseptic, and photo-protected aqueous steady state.

→By in vitro standards, these conditions appear mild.

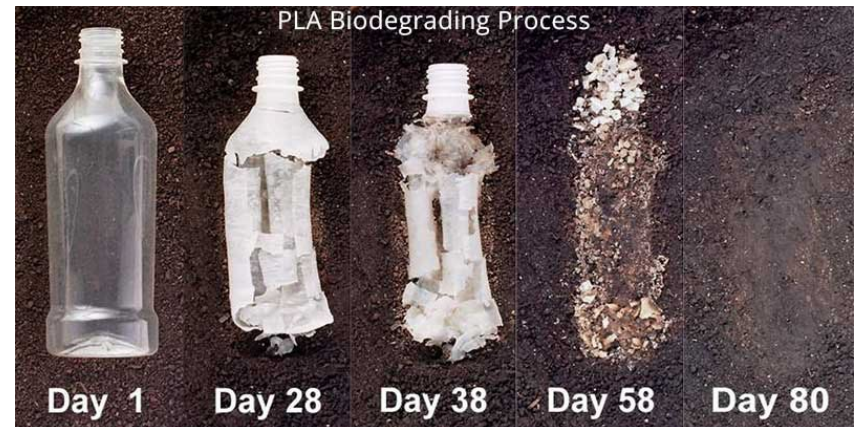
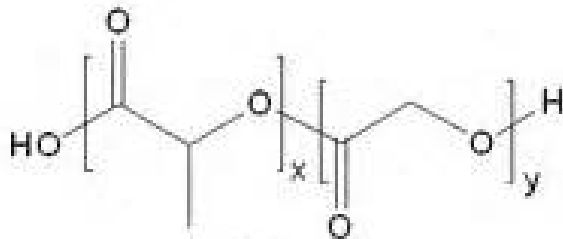
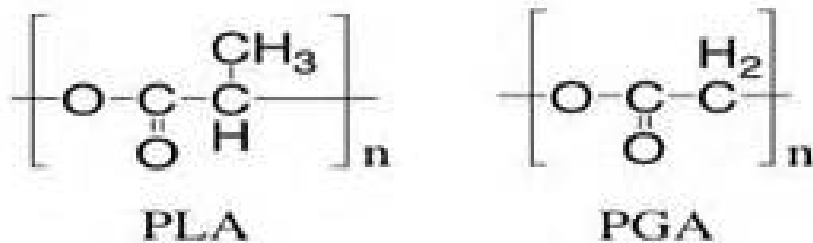
→ However, complex interactions of humoral and cellular components of body fluids involving activators, receptors, inhibitors, etc. produce aggressive responses to any foreign bodies through the processes of adhesion, chemical reaction, and particulate transport.

Host-induced Hydrolytic Processes

Several scenarios leading to hydrolysis in the host can be considered.

→ First, essentially neutral water is capable of hydrolyzing certain polymers (e.g., polyglycolic acid and PGA) at a significant rate.

→ However, this simple mechanism is unlikely to be significant in polymer compositions selected for long-term in vivo biostability.



a PLA plastic bottle on the ground, which was shown to be degradable in 80 days

<http://2016.igem.org/Team:Evry/HP/Gobelets>

Host-induced Hydrolytic Processes

Next, ion-catalyzed hydrolysis offers a likely scenario in body fluids.

→ Extracellular fluids contain ions such as: H^+ , OH^- , Na^+ , Cl^- , HCO_3^- , PO_4^{3-} , K^+ , Mg^{+2} , Ca^{+2} , and SO_4^{2-} . Organic acids, proteins, lipids, lipoproteins, etc. also circulate as soluble or colloidal components.

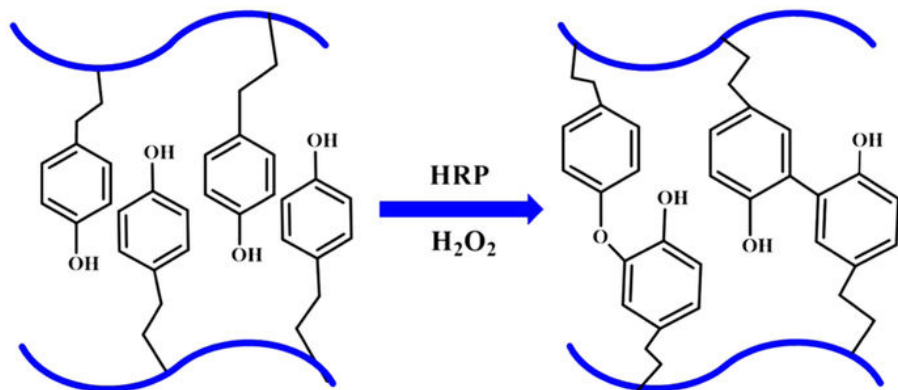
→ It has been shown that certain ions (e.g., PO_4^{3-}) are effective hydrolysis catalysis, enhancing reaction rates of polyesters by several orders of magnitude.

→ Ion catalysis may be a surface effect or a combined surface bulk effect, depending on the hydrophilicity of the polymer.

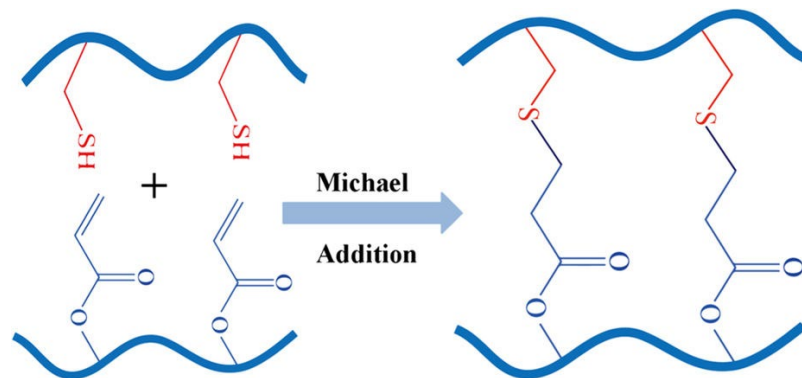
→ Very hydrophobic polymers (e.g., those containing <2% water of saturation) absorb negligible concentrations of ions.

→ **Hydrogels**, on the other hand, which can absorb large amounts of water (>15% by weight) are essentially "sieves," allowing significant levels of ions to be absorbed with consequent bulk hydrolysis via acid, base, or salt catalysis.

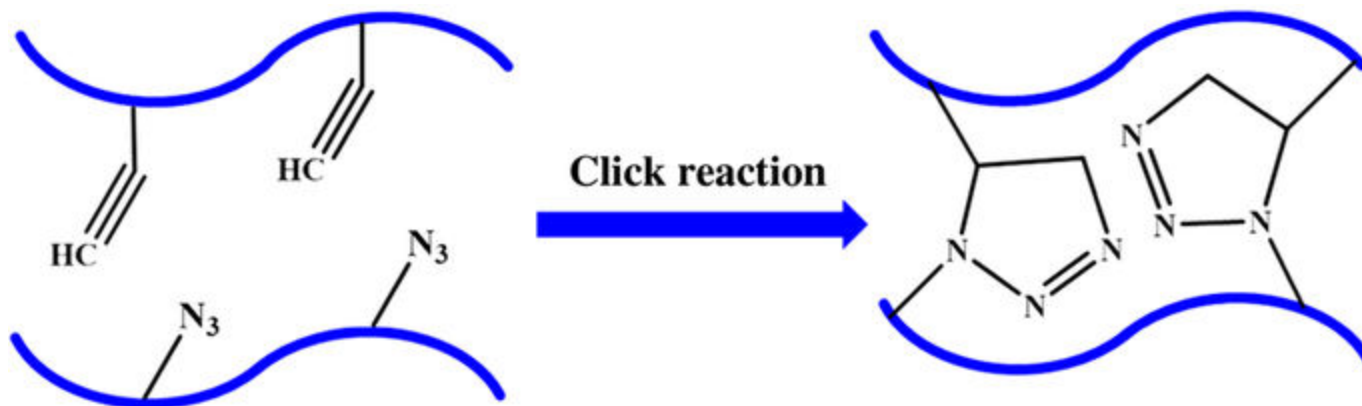
Recall the preparation of hydrogels (a few approaches)



Schematic illustration of injectable hydrogels prepared by the enzymatic cross-linking method with horseradish peroxidase (HRP) and H₂O₂.

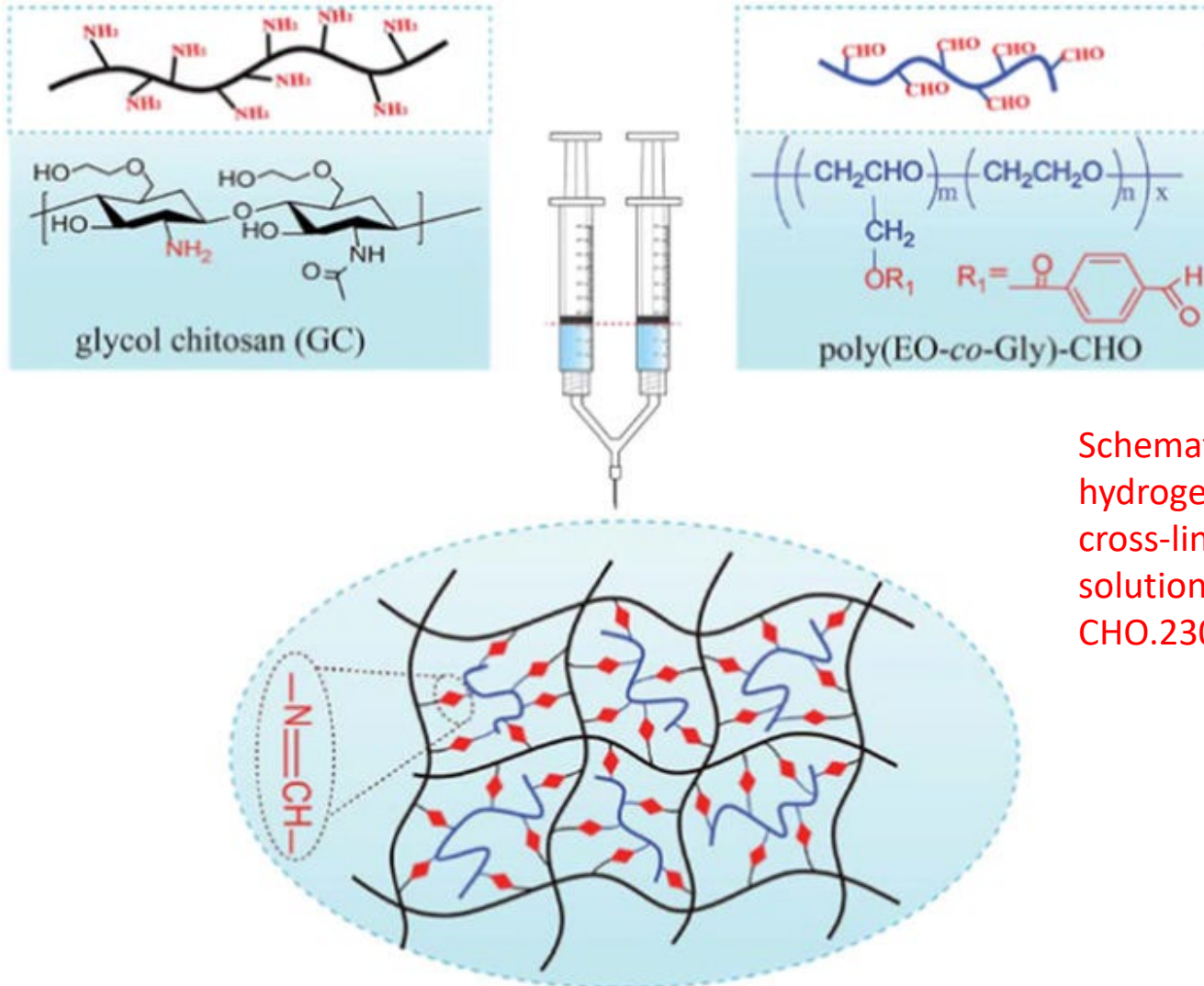


Schematic illustration of injectable hydrogels prepared by the Michael addition cross-linking method.



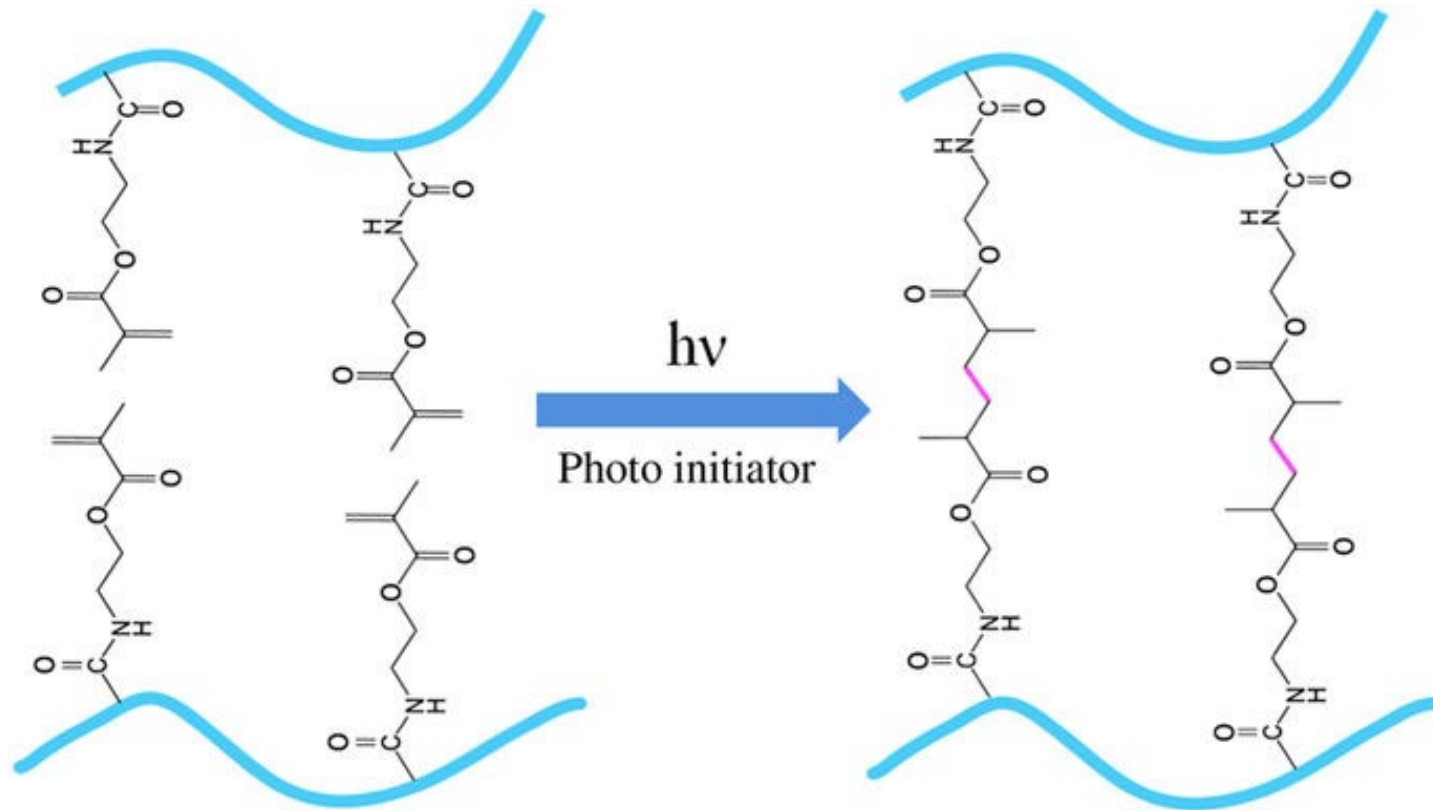
Schematic illustration of injectable hydrogels prepared by click chemistry.

Recall the preparation of hydrogels (a few approaches)



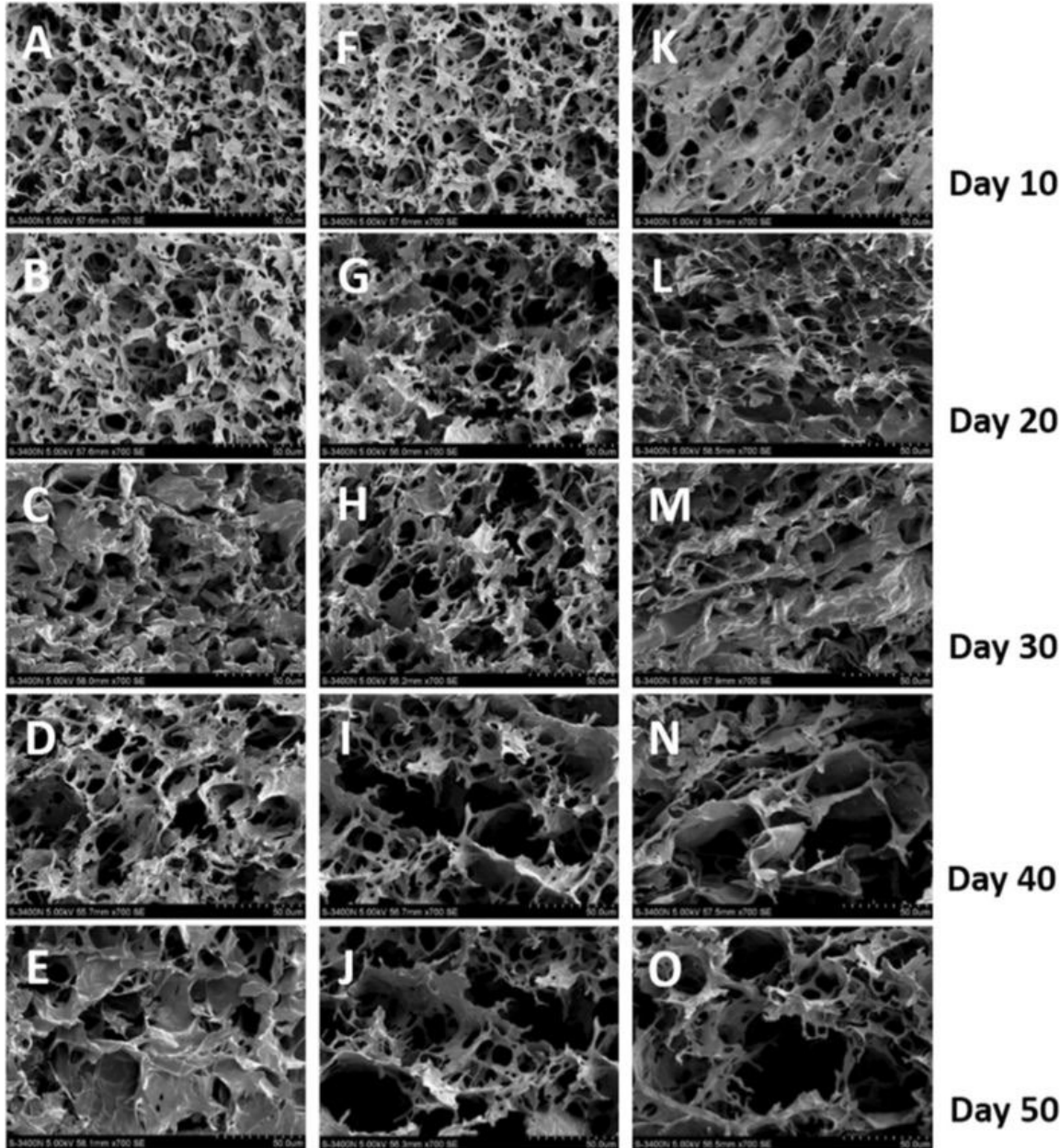
Schematic illustration of injectable hydrogels prepared by Schiff base cross-linking between aqueous solutions of GC and poly(EO-co-Gly)-CHO.230

Recall the preparation of hydrogels (a few approaches)



Schematic illustration of injectable hydrogels prepared by the photo-cross-linking method

Host-induced Hydrolytic Processes



FE-SEM images of hydrogel. Hydrogel incubated in the absence (A-E) or presence of, 0.1 g/L and 0.5 g/L lysozyme (F-J and K-O, respectively; scale bar = 50 μm).

<https://jnanobiotechnology.biomedcentral.com/articles/10.1186/1477-3155-12-23>

Host-induced Hydrolytic Processes

Localized pH changes in the vicinity of the implanted device, which usually occur during acute inflammation or infection, can cause catalytic rate enhancement of hydrolysis.

Enzymes generally serve a classic catalytic function, altering reaction rate (via ion or charge transfer) without being consumed, by modifying activation energy but not thermodynamic equilibrium.

→ While enzymes function in extracellular fluids, they are most effectively transferred onto target substrates by direct cell contact (e.g., during phagocytosis).

→ Hydrolytic enzymes or hydrolases (e.g., proteases, esterases, lipases, glycosidases) are named for the molecular structures they affect. They are cell-derived proteins which act as highly specific catalysis for the scission of water-labile functional groups.

Host-induced Hydrolytic Processes

Enzymes contain molecular chain structures and develop conformations that allow "recognition" of chain sequences (receptors) on biopolymers.

→Lacking the recognition sequences of susceptible natural polymers, most synthetic polymers are more resistant to enzymatic degradation.

→Nevertheless, comparative studies have shown some enhancement of hydrolysis rates by enzymes, particularly with synthetic polyesters and polyamides.

→Apparently the enzymes can recognize and interact with structural segments of the polymers or more accurately, of the polymers coated with serum proteins, to initiate their catalytic action in vitro.

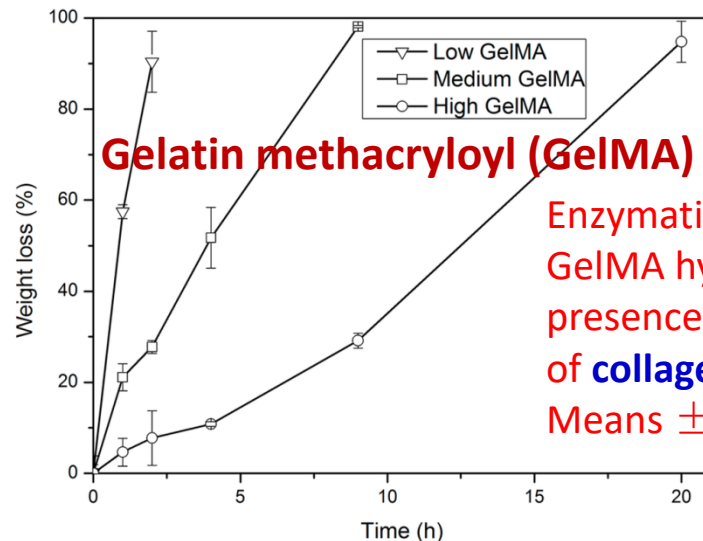
Host-induced Hydrolytic Processes

Enzymes with demonstrated effects on hydrolysis rates can be quite selective in the presence of several hydrolyzable functional groups.

→For example, polyether urethane and polyester urethane exposed to **hydrolytic enzymes** (an esterase, cholesterol esterase, and a protease, elastase) were observed for rate and site of hydrolysis.

→**Enzyme catalysis** was clearly observed for the ester groups while the hydrolytically susceptible urea, urethane, and ether groups did not show significant hydrolysis as indicated by release of radio-labeled degradation products.

Implanted devices that are in continuous motion relative to neighboring tissue can provoke inflammation, stimulating enzyme release.



| Gelatin Methacryloyl (GelMA)

GelMA作为无菌冻干产品提供，在PBS或HEPES缓冲液中以适合您的应用的浓度复溶，并将其与光引发剂混合，使您的水凝胶在体温下可光交联且稳定。

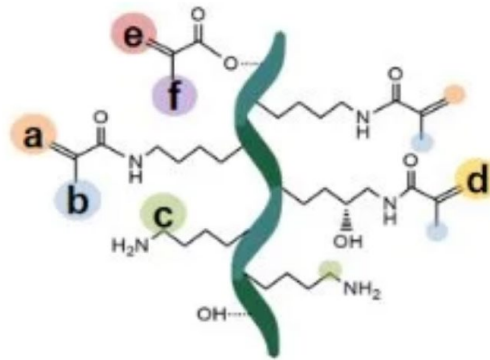
由于其无与伦比的可调性和生物活性，GelMA作为生物材料在组织工程、3D生物打印和3D细胞培养应用中非常受欢迎。可以单独使用，也可以与其他材料混合，以创建可打印细胞外基质。

应用

猪皮凝胶MA

牛骨凝胶MA

冷水鱼皮凝胶MA



Low endotoxin gelatin, type A is sterile filtrated through 0.2 μm sterile filter, and lyophilized for use in biomedical applications.

Gelatin is widely used for tissue engineering and 3D bioprinting. Gelatin is derived from natural extracellular matrix (ECM) components. Due to its low cost, abundance, and retention of natural cell binding motifs, gelatin has become a highly sought material for tissue engineering applications. Gelatin solution has thermoreversible gelling property which enables synthesis of biocompatible and biodegradable hydrogels and promote cell adhesion, spreading, and proliferation.

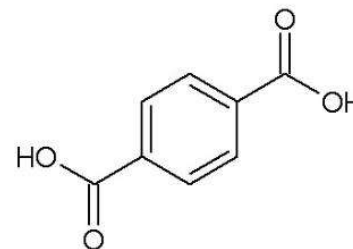
Hydrolysis: Pre-clinical and Clinical Experience

Polyester

Poly(ethylene terephthalate) (PET, $(-O-C-C-O-CO-X-CO-)_n$, in woven, velour, or knitted fiber configurations, remains a primary choice of cardiovascular surgeons for large-diameter vascular prostheses, arterial patches, valve sewing rings, etc.

→It is a strong, flexible, polymer, stabilized by high crystallinity as a result of chain rigidity and orientation and is often considered to be bio-stable.

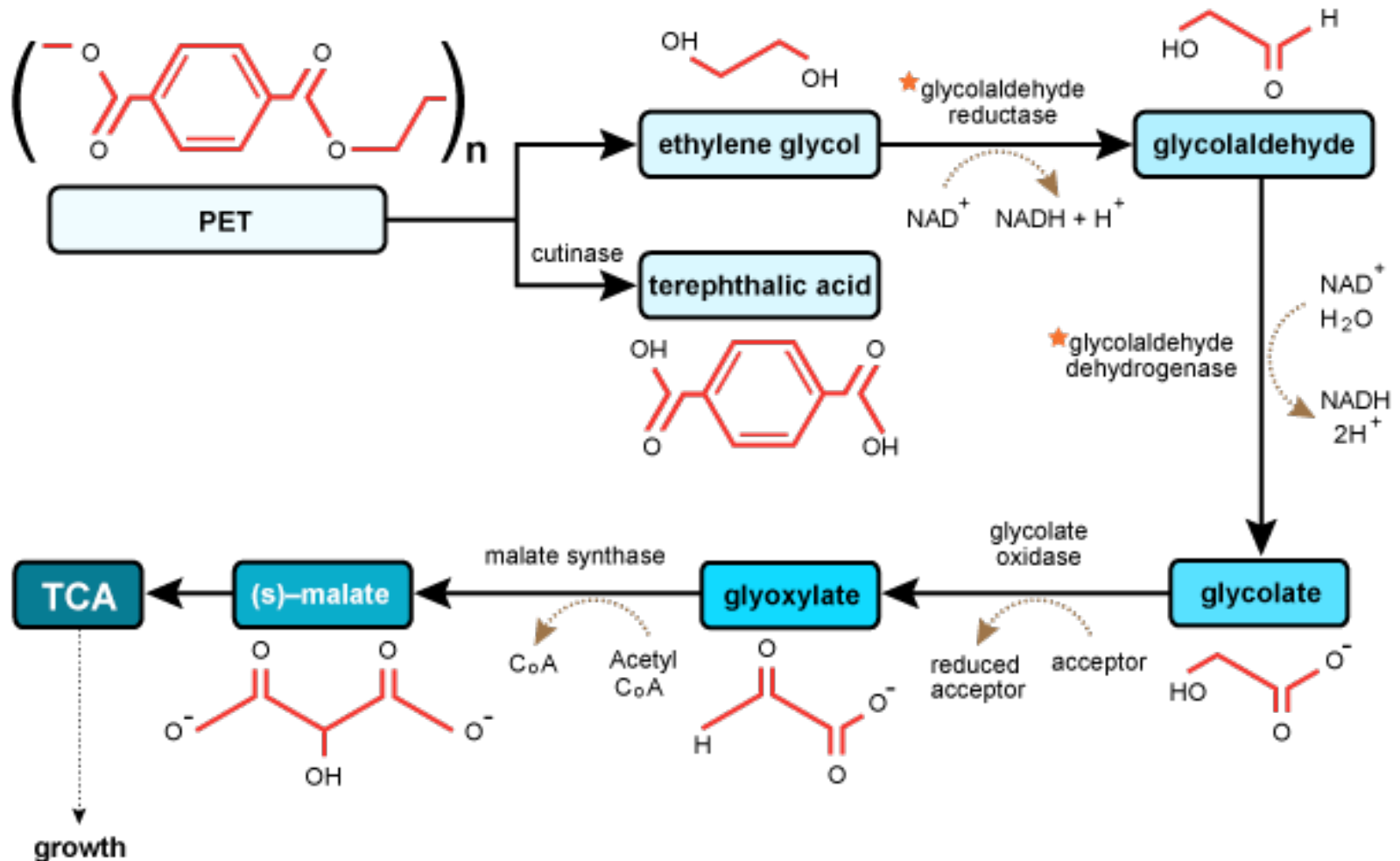
→Yet, over several decades, there have been numerous reports of long-term degeneration of devices in vivo, owing to breakage of fibers and device dilation.



Phthalic acid

Do you remember the PET structure?

Hydrolysis: Pre-clinical and Clinical Experience



http://2012.igem.org/Team:UC_Davis/Project/Catalyst

Hydrolysis: Pre-clinical and Clinical Experience

Systematic studies of PET implants in healthy dogs have shown slow degradation rates, which were estimated to be equivalent to those in human.

→For woven patches implanted subcutaneously, a mean total time by the body of 30 ± 7 years, with 50% deterioration of fiber strength in 10 ± 2 years was projected.

→In infected dogs, however, where localized pH dropped to as low as 4.8, degradation was enhanced exponentially.

Aliphatic polyesters are most often intended for use as biodegradable polymers, with polycaprolactone, for example, undergoing a significant **decrease in molecular weight** as indicated by a drop of 80%-90% in relative viscosity within 120 weeks of implant.

Polyester Urethane

The earliest reported implants of polyureathanes, dating back to the 1950s, were cross-linked, aromatic polyester urethane foam compositions.

→ Their use in plastic and orthopedic reconstructive surgery initially yielded promising results.

→ acute inflammation was low.

→ Tissue ingrowth promoted thin fibrous capsules.

→ However, within months, they were degraded and fragmented, producing untoward chronic effects.

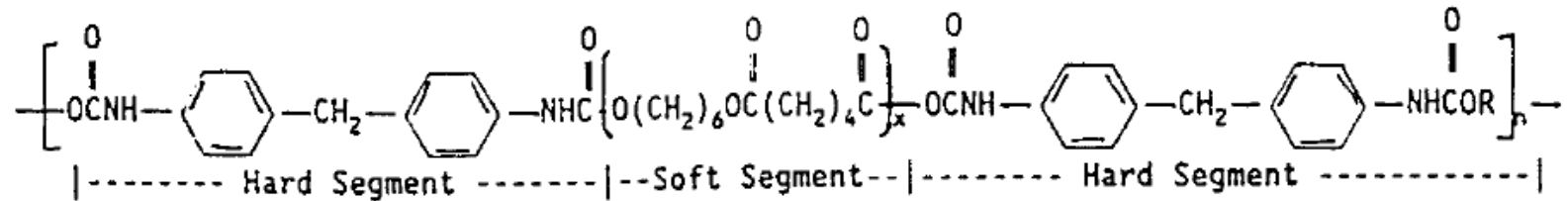
→ initial degradation are generally considered to be the polyadipate ester soft segments which undergo hydrolysis.

Polyester Urethane

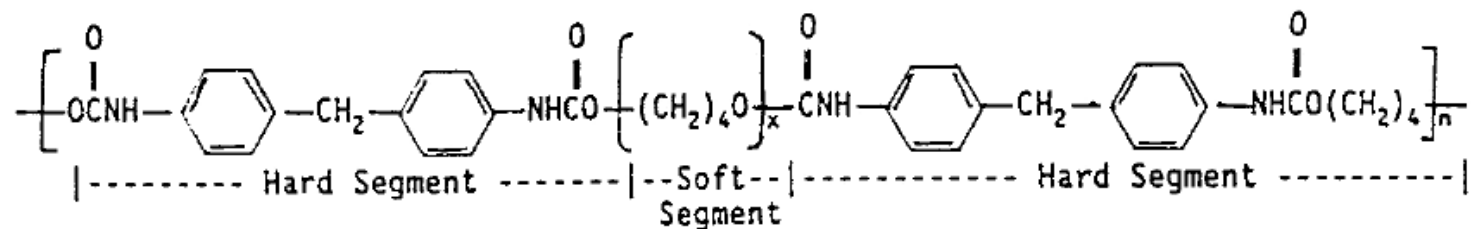
By comparison, corresponding polyether urethanes are very resistant to hydrolysis, although more susceptible to oxidation.

However, all these polymers subsequently produce meaningful levels of aromatic amines (suspected carcinogene) after degraded in vivo. It is currently an unresolved subject of considerable debate.

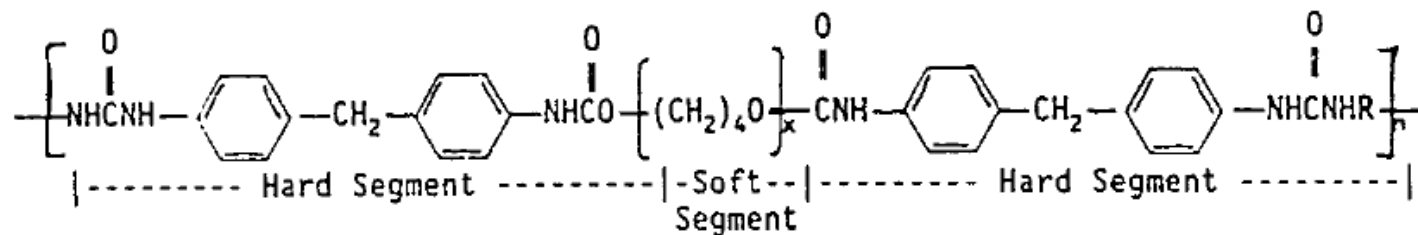
Polyester Urethane



Polyester urethane (e.g., Surgitek, Meme Mammary Prosthesis Covering)



Polyether urethane (eg., Dow Pellethane 2363 Series)



Polyether urethane urea (eg., Ethicon Biomer)

FIG. 3. Structure of implantable polyester urethane, polyether urethane, and polyether urethane urea.

Polyamides

Nylon 6 (polycaproamide) and nylon 66 (poly(hexamethylene adipamide)) contain a hydrolyzable amide connecting group, as do proteins.

→these synthetic polymers can absorb 9-11% water, by weight, at saturation.

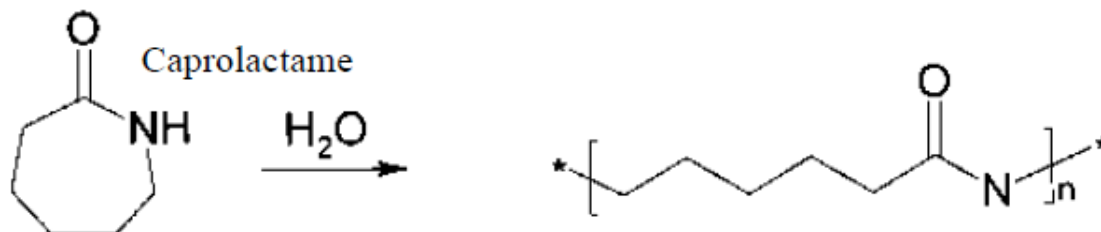
→it is predictable that they degrade by ion-catalyzed surface and bulk hydrolysis.

→in addition, hydrolysis due to enzymatic catalysis leads to surface erosion.
(25%/89days, 83%/726days)

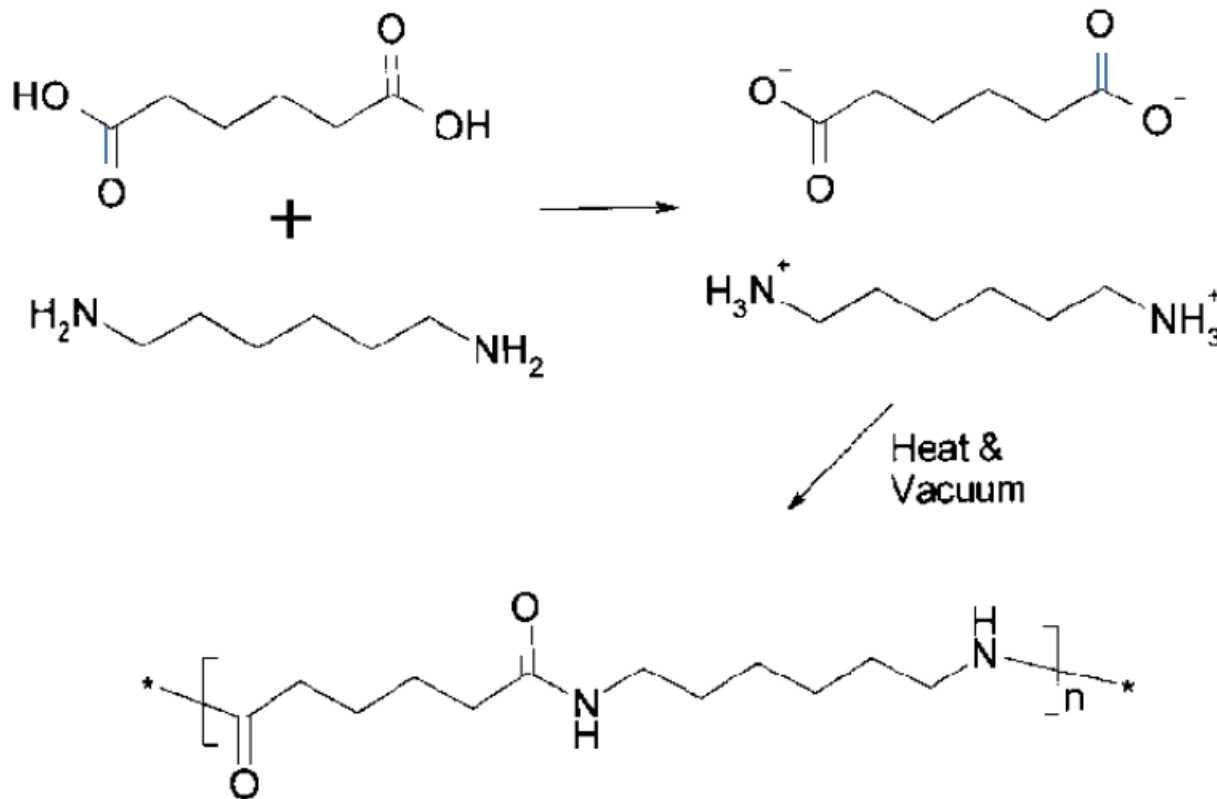
→polyamides with long aliphatic hydrocarbon chain segments are more hydrolytically stable than shorter chain nylons and correspondingly degrade slower in vivo.

Polyamides

Nylon 6

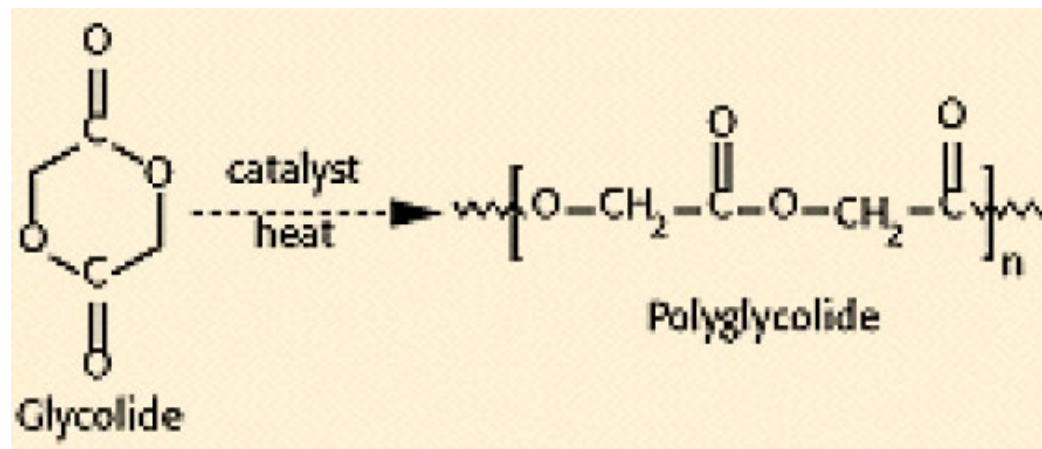


Nylon 6,6

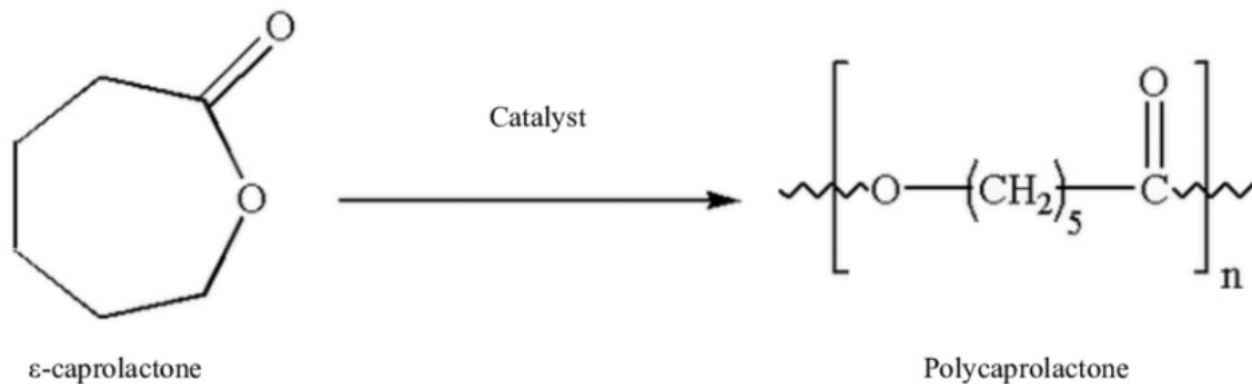


Polyglycolide (PGA). Polyglycolide is the simplest linear aliphatic polyester. PGA was used to develop the first totally synthetic absorbable suture, marketed as Dexon in the 1960s by Davis and Geck, Inc. (Danbury, CT).

Glycolide monomer is synthesized from the dimerization of glycolic acid. Ring-opening polymerization yields high-molecular-weight materials, with approximately 1—3% residual monomer present (see Figure 1). PGA is highly crystalline (45—55%), with a high melting point (220—225°C) and a glass transition temperature of 35—40°C. Because of its high degree of crystallization, it is not soluble in most organic solvents; the exceptions are highly fluorinated organics such as hexafluoroisopropanol. Fibers from PGA exhibit high strength and modulus and are too stiff to be used as sutures except in the form of braided material. **Sutures of PGA lose about 50% of their strength** after 2 weeks and 100% at 4 weeks, and are completely absorbed in 4—6 months. Glycolide has been copolymerized with other monomers to reduce the stiffness of the resulting fibers.



Poly(ϵ -caprolactone). The ring-opening polymerization of ϵ -caprolactone yields a semicrystalline polymer with a melting point of 59—64°C and a glass-transition temperature of -60°C . The polymer has been regarded as tissue compatible and used as a biodegradable suture in Europe. Because the homopolymer has a degradation time on the order of 2 years, copolymers have been synthesized to accelerate the rate of bioabsorption. For example, **copolymers of ϵ -caprolactone with dl-lactide have yielded materials with more-rapid degradation rates.** A block copolymer of ϵ -caprolactone with glycolide, offering reduced stiffness compared with pure PGA, is being sold as a monofilament suture by Ethicon, Inc. (Somerville, NJ), under the trade name Monacryl.



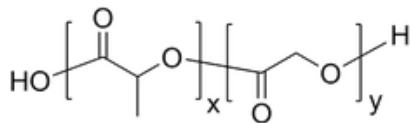
Synthesis of poly(ϵ -caprolactone).

Poly(lactide-co-glycolide). Using the polyglycolide and poly(l-lactide) properties as a starting point, it is possible to copolymerize the two monomers to extend the range of homopolymer properties.

Copolymers of glycolide with both l-lactide and dl-lactide have been developed for both device and drug delivery applications. It is important to note that there is not a linear relationship between the copolymer composition and the mechanical and degradation properties of the materials. For example, a copolymer of 50% glycolide and 50% dl-lactide degrades faster than either homopolymer. Copolymers of l-lactide with 25—70% glycolide are amorphous due to the disruption of the regularity of the polymer chain by the other monomer. A copolymer of 90% glycolide and 10% l-lactide was developed by Ethicon as an absorbable suture material under the trade name Vicryl. It absorbs within 3—4 months but has a slightly longer strength-retention time.

PLGA (50:50) (poly(lactic-co-glycolic acid) (50:50)) 是聚乳酸 (PLA) 和聚乙醇酸 (PGA) 的共聚物, 其可制造用于药

MCE 的所有产品仅用作科学研究或药证申报, 我们不为任何个人用途提供产品和服务



规格	价格	是否有货	数量
500 mg	¥ 650	In-stock	0
1 g	¥ 1100	In-stock	0
5 g	询价		
10 g	询价		

Oxidative Biodegradation

Oxidation Reaction Mechanisms and Polymer Structures

While much is known about the structures and reaction products of polymers susceptible to **oxidative biodegradation**, confirmation of the individual reaction steps has not yet been demonstrated analytically.

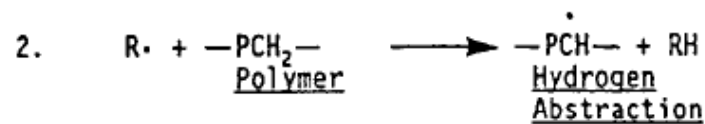
→The polymer oxidation processes to be discussed may be consistent with a **homolytic chain reaction or a heterolytic** mechanism.

→Species such as carbonyl, hydroxyl, and chain scission products are detectable. Classic initiation, propagation, and termination events for homolysis and ionic heterolytic processes are detailed in Fig. 4.

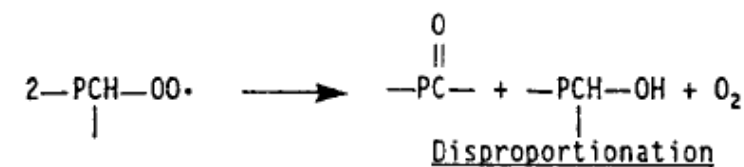
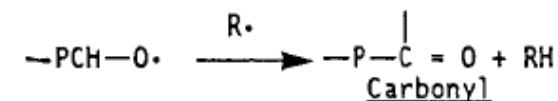
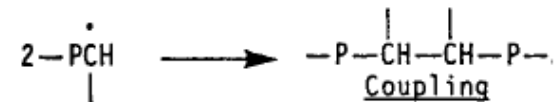
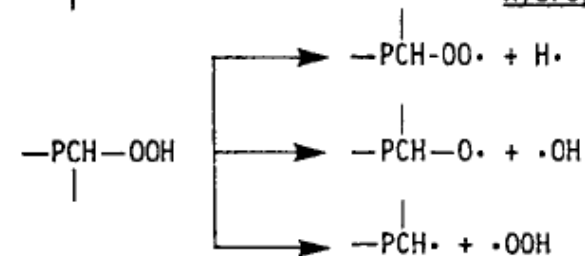
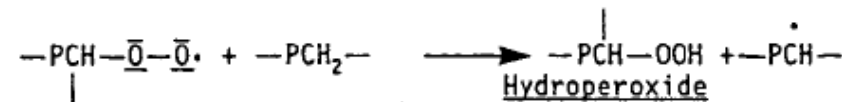
Homolysis



Initiation



Propagation



Termination

Figure 4a

Heterolysis

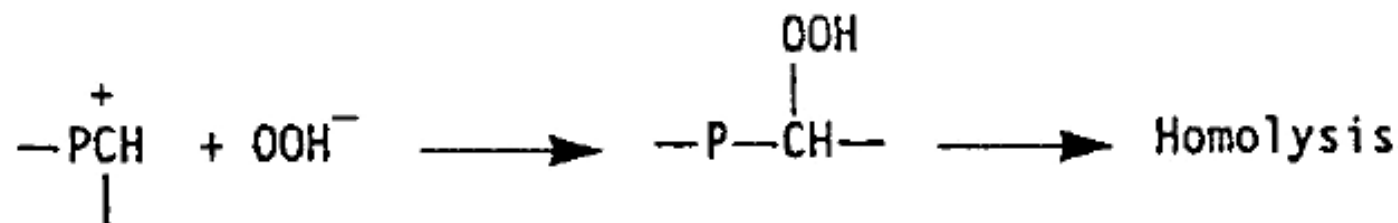
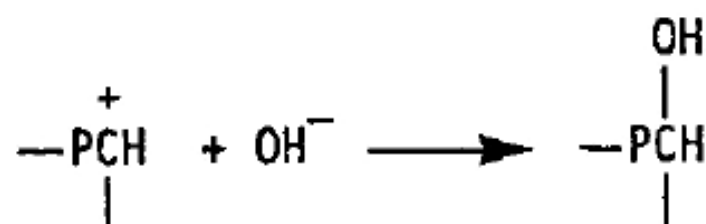
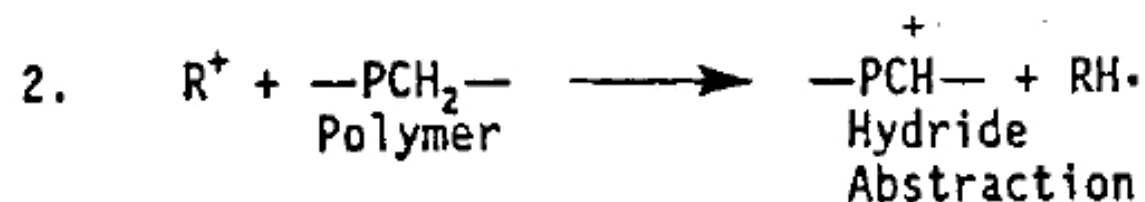
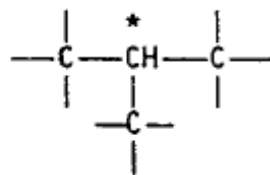
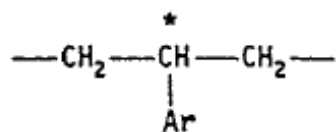


FIG. 4. Proposed homolytic chain reaction and heterolytic oxidation mechanisms.

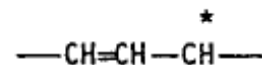
A selection of readily oxidized groups and the atom at which initial attack occurs.



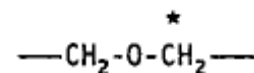
Branched Aliphatic Hydrocarbon



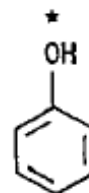
Ar = Aromatic Ring-Containing Polymer



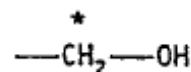
Allylic Hydrocarbon



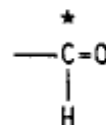
Ether



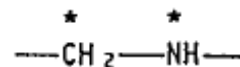
Phenol



Alcohol

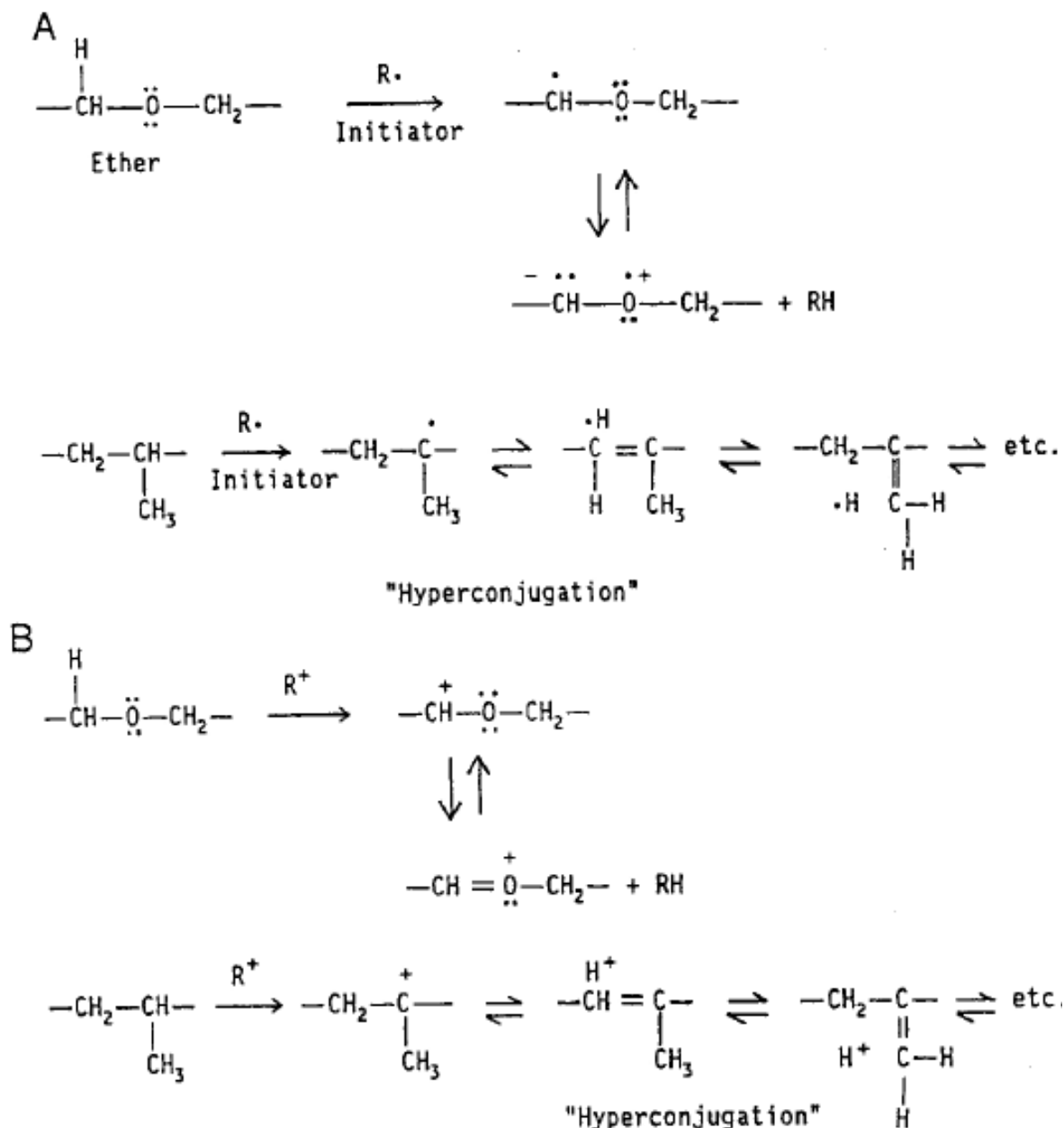


Aldehyde



Amine

FIG. 5. Readily oxidizable functional groups (* is site of homolysis or heterolysis).



In Fig. 6, examples of radical and ion stabilization by resonance in ether and branched hydrocarbon structures are provided. Peroxy, carbonyl, and other radical intermediates are stabilized by similar resonance delocalization of electrons from the elements C, O, H, or N.

FIG. 6. (A) Resonance stabilization of ether and hydrocarbon radicals. (B) Resonance stabilization of ether and hydrocarbon cations.

Direct Oxidation by Host

In these circumstances, host-generated molecular species effect or **potential oxidative processes** directly on the polymer.

→Based on solid analytical evidence, is that such reactive molecules are derived from activated **phagocytic cells** responding to the injury and properties of the foreign body at the implant site

→These cells, which originate in the bone marrow and populate the circulatory system and connective tissues, are manifest as two types, the **neutrophils (polymorphonuclear leukocytes, PMNs)** and the **monocytes**.

→The latter can differentiate into **macrophage and foreign body giant cell (FBGC)** phenotypes.

Direct Oxidation by Host

Much work is under way to elucidate the sequence of events leading to phagocytic oxidation of biomaterials.

→ Certain important processes of wound healing in the presence of biologically derived foreign bodies such as bacteria and parasites, are showing some relevance to biomaterial implants.

Neutrophils, responding to chemical mediators at the wound site, mount a powerful but transient chemical attack within the first few days of injury.

→ Chemically susceptible biomaterials may be affected if they are in close apposition to the wound site.

→ Activated macrophages subsequently multiply and subside within days at a benign wound site or in weeks if stimulants such as toxins or particulates are released at the site. Their fusion products, foreign body giant cells, can survive for months to years on the implant surface. Macrophages also remain resident in collagenous capsules for extended periods.

The following discussion attempts to provide logical biological pathways to powerful oxidants capable of producing known degradation products.

→Both PMNs and macrophages metabolize oxygen to **form a superoxide anion ($O_2^{\cdot-}$)**.

→This intermediate can undergo transformation to **more powerful oxidants**, or, conceivably, can initiate **homolytic reactions** on the polymer.

→Superoxide dismutase (SOD), a **ubiquitous peroxidase enzyme**, can catalyze the conversion of superoxide to hydrogen peroxide, which, in the presence of myeloperoxidase (MPO), derived from PMNs, is converted to **hypochlorous acid (HOCl)**.

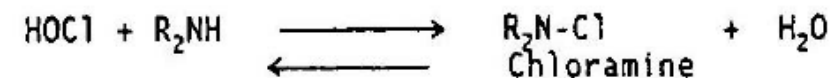
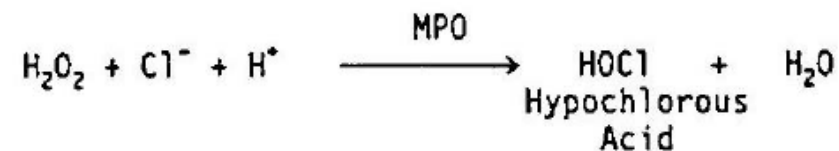
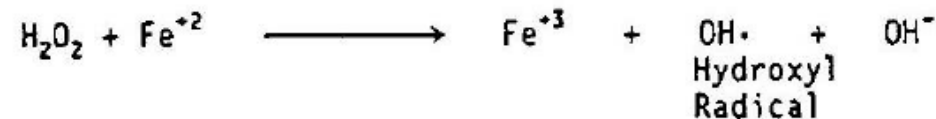
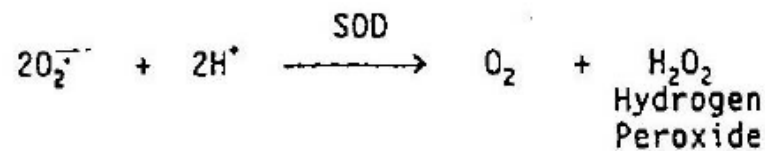
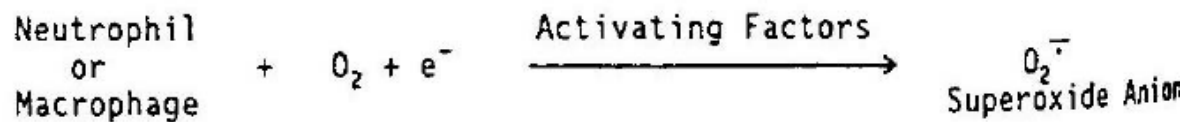
→A potent biomaterial oxidant in its own right, hypochlorite can oxidize free amine functionality (e.g., in proteins) to chloramines that can perform as long-lived sources of chlorine oxidant. Hypochlorite can oxidize other substituted nitrogen functional groups (amides, ureas, urethanes, etc.) with potential chain cleavage of these groups.

The following paragraphs describe potential cooperative reactions involving acquired peroxidase and free ferrous ions.

→ Macrophages contain essentially no myeloperoxidase (MPO) , so their hydrogen peroxide is not normally converted to HOCl.

→ However, PMN-derived MPO can bind securely to foreign body surfaces, and serve as a catalyst reservoir for macrophage- or FBGC-derived HOCl production.

→ If free ferrous ion, which is normally present in negligible quantities in the host, is released to the implant site by hemolysis or other injury, it can catalyze the formation of the powerfully oxidizing hydroxyl radical (Fig.7).

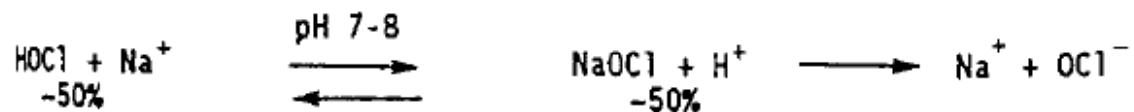


Superoxide dismutase (SOD),

myeloperoxidase (MPO)

FIG. 7. Generation of potential oxidants by phagocytic processes.

Equilibrium Products



Radical Intermediates

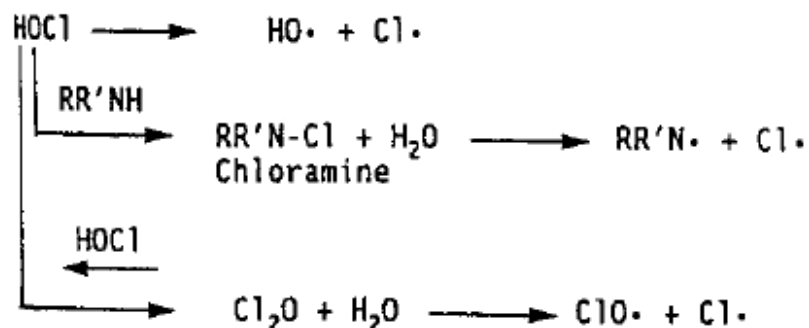


Figure 8 shows radical and ionic intermediates of HOCl that may initiate biomaterial oxidation.

Ionic Intermediates

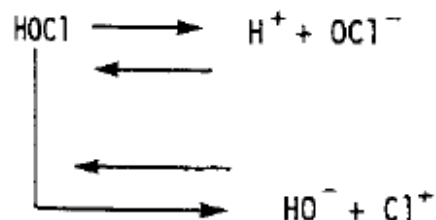
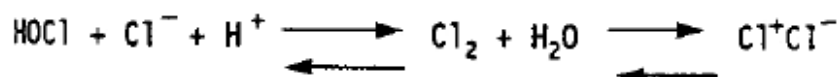


FIG. 8. Hypochlorous acid: formation and potential reaction intermediates.

Stress Cracking

Stress cracking of polyurethane is characterized by surface attack of the polymer by chemical changes induced by relatively specific *in vivo* or *in vitro* oxidizing conditions.

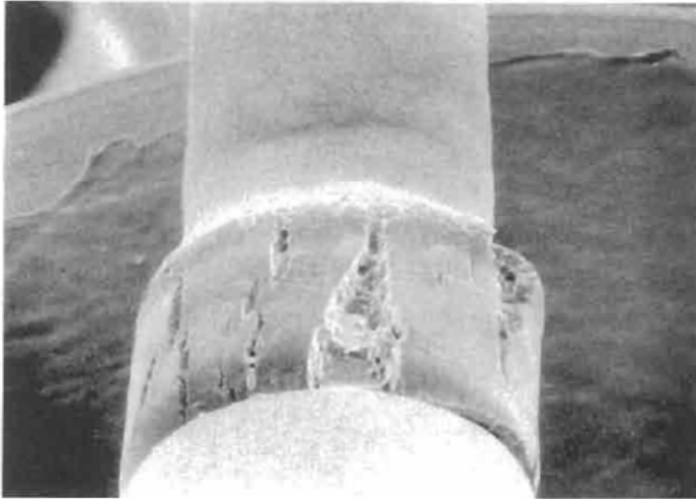


FIG. 11. Pellethane 2463-80A tubing with high applied radial stress showing total breach.

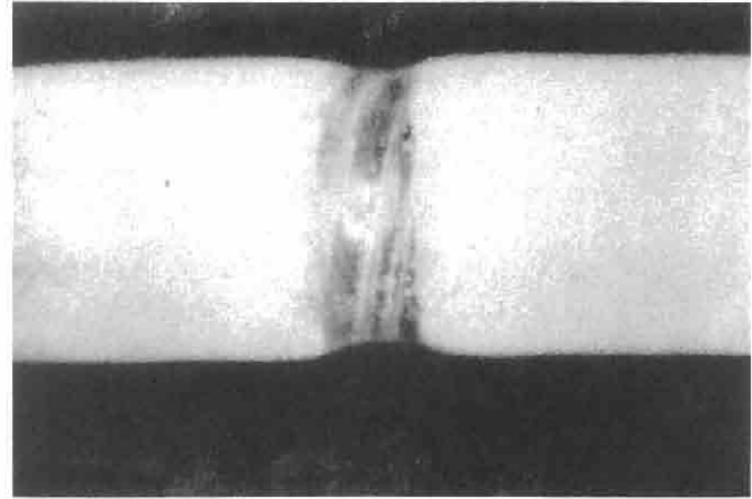


FIG. 12. Pellethane 2363-80A tubing showing "frosting" due to stress from tight ligature.

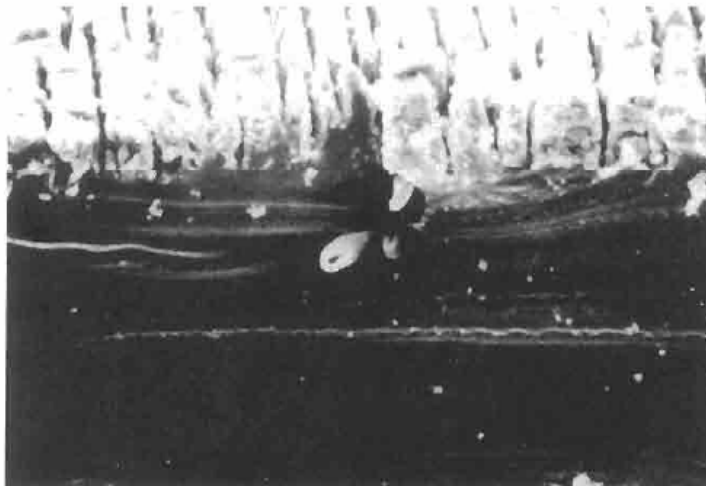


FIG. 13. Stress crack pattern (frosting) near tight ligature, $\times 14$.

Device or Environment-mediated Oxidation

Metal Ion-induced Oxidation

Metal ion-induced oxidation, has been confirmed by in vitro studies in which polyether urethanes were aged in metal ion solutions of different standard oxidation potentials. Above an **oxidation potential** of about +0.77, chemical degradation was severe. Below that oxidation potential, changes in the polymer that are characteristic of simple plasticization were seen.

→ This technique also showed that metal ion-induced oxidation was proportional to the ether content of the polyurethane

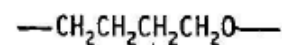
→ The metal ion-induced oxidation **process clearly involves corrosion of metallic** elements to their ions and subsequent oxidation of the polymer.

→ In operating devices, the metal ion may be formed by solvation, galvanic corrosion, or chemical or biochemical oxidation. In turn, these metal ions develop oxidation potentials that may well be enhanced in body fluids over their standard half-cell potentials.

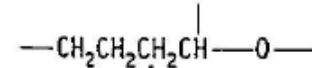
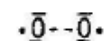
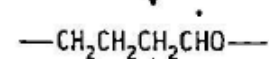
→ As strong oxidants, **they produce intermediates or attack the polymer to initiate the chain reaction.**

→ Metal ion-induced oxidation is, therefore, the result of a highly complex interaction of the device, the polymer, and the body.

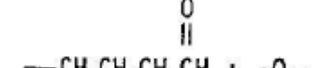
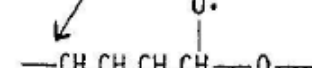
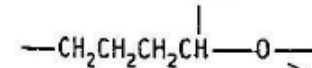
Homolysis



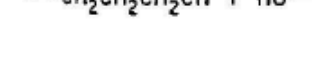
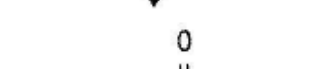
A• or (X)



RH

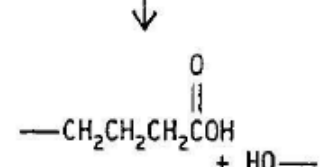
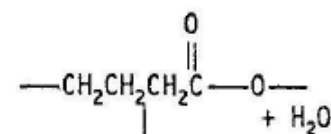


RH



A• = OH•, O₂•, Cl•
RR'N•, OOH•, etc.

(X) = Mⁿ⁺,
hν, Δ, etc.



Heterolysis

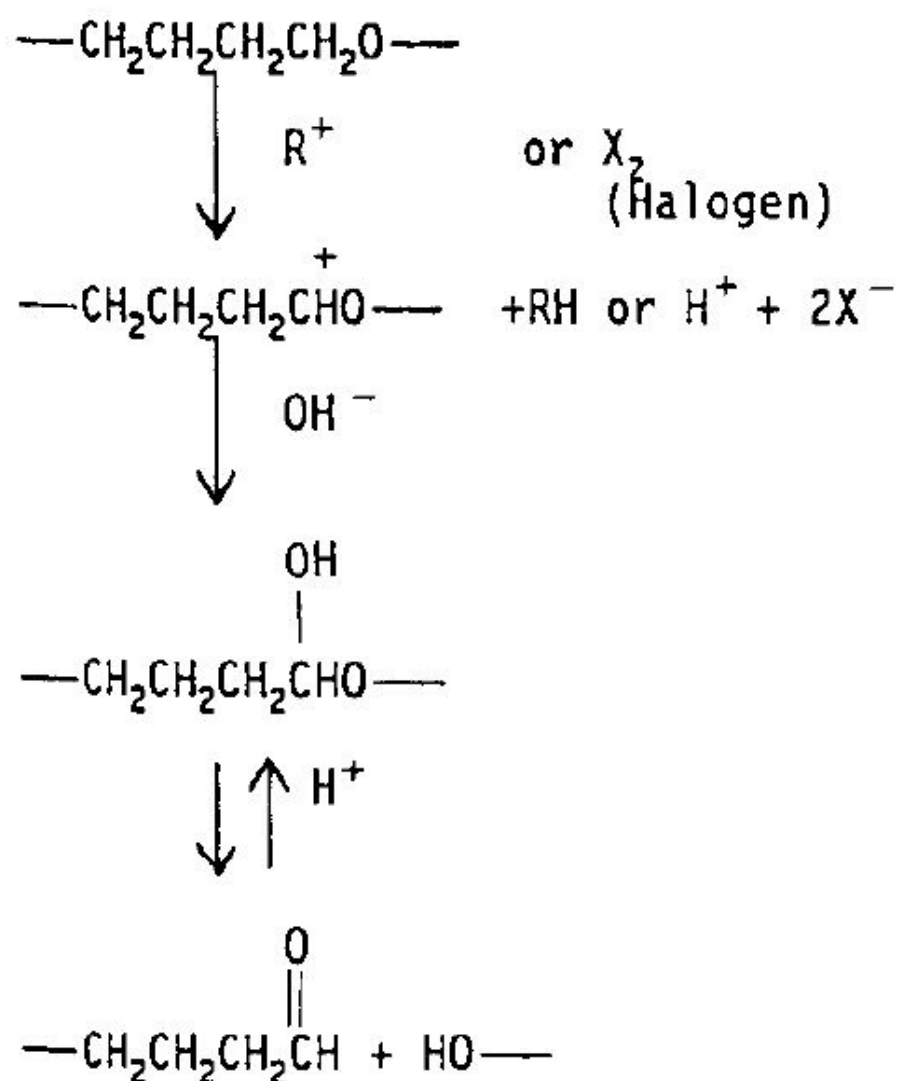


FIG. 15. Pathways for oxidative fragmentation of polyethers.

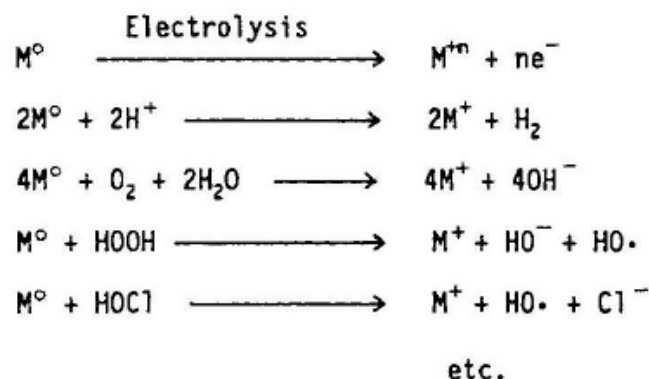


FIG. 19. Formation of metal ion from metal.

CONCLUSION

Polymers that are carefully chosen for use in implanted devices generally serve effectively for their intended lifetimes if they are properly processed and device-material-host interactions are adequately addressed. In certain limited circumstances, unintended hydrolytic or oxidative biodegradation occurs. This may be induced by direct attack by the host or via the intermediacy of the device or the outside environment. With susceptible polymers, protective measures can be taken to ensure extended efficacy, although new, biodegradation-resistant polymers, which are on the horizon, will require less protection. Knowledge of biodegradation mechanisms and the employment of appropriate countermeasures will promote the continued growth in compositions and uses of polymers as implantable biomaterials.

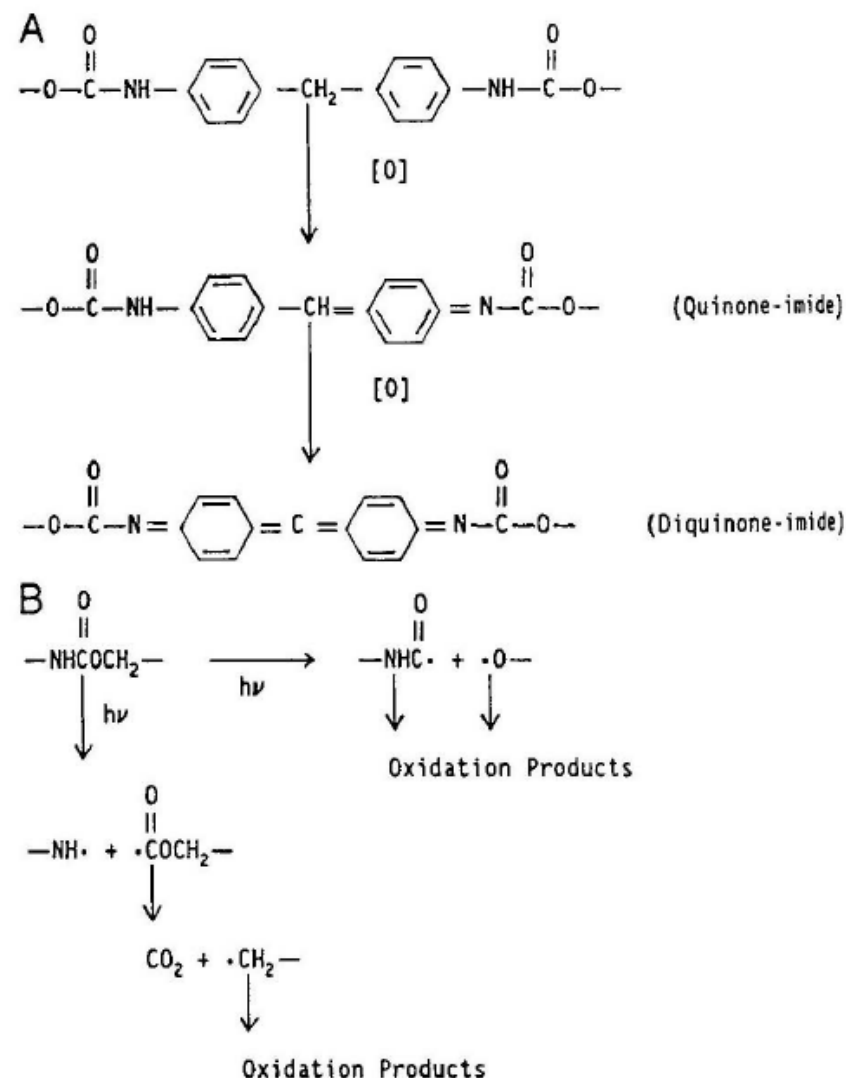


FIG. 21. Photo-oxidative reactions of aromatic polyurethanes. (A) Formation of quinone-imide from aromatic polyurethane. (From A. J. Coury *et al.*, *J. Biomater. Appl.* 3, 1988.) (B) Photolytic cleavage of urethane link. (From S. K. Brauman *et al.*, *Ann. Biomed. Eng.* 9, 1981.)

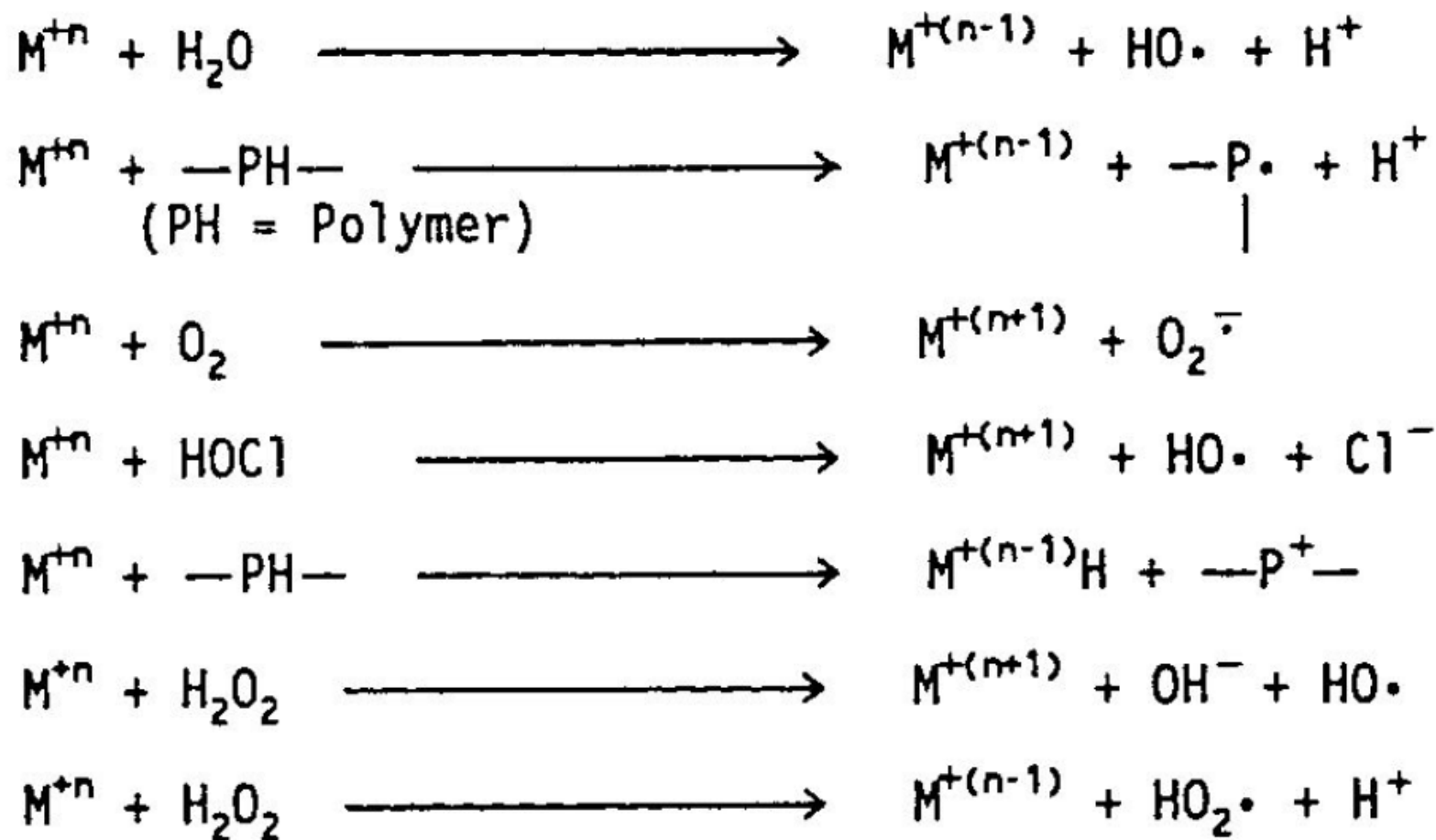


FIG. 20. Initiation of oxidation pathways by metal ion.

Oxidative Degradation Induced by External Environment

Under very limited circumstances the body can transmit electromagnetic radiation that may affect the integrity of implanted polymers.

→ For example, the cornea and vitreous humor of the eye as well as superficial skin layers allow the passage of long-wave (320-400 nanometer) “ultraviolet A” radiation.

→ Absorption of **ultraviolet radiation** causes electron excitation that can lead to photo-oxidative degradation. This process has been suggested in the **breakdown of polypropylene components of intraocular lens**.



Figure 16. Random crack pattern of Pellethane 2363-80A lead insulation caused by metal ion-induced oxidation x 480.



Figure 18. Crack pattern on inner lumen of polyether urethane lead insulation tracking coil indicative of metal ion induced oxidation. x100.

Degradative effects of the biological environment on metals

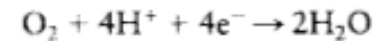
The most pertinent form of corrosion related to metallic biomaterials is aqueous corrosion. This occurs when electrochemical reactions take place on a metallic surface in an aqueous electrolyte. There are always two reactions that occur: the anodic reaction, which yields metallic ions, for example, involving the oxidation of the metal to its salt:



and the cathodic reaction, in which the electrons so generated are consumed. The precise cathodic reaction will depend on the nature of the electrolyte, but two of the most important in aqueous environments are the reduction of hydrogen:



and the reduction of dissolved oxygen:



in acidic solutions or:

